David Xia

davidx3@illinois.edu | Personal Website | LinkedIn - dx3 | GitHub - davidxia3

Research Interests

Theoretical ML, reinforcement learning, online learning, MDP function approximation, sample complexity

Education

- **University of Illinois Urbana-Champaign** *Aug 2023 - May 2027*
  Bachelor of Science in Computer Science Champaign, Illinois
  Bachelor of Science in Liberal Arts & Sciences Major in Mathematics, Data Optimization Concentration
  Bachelor of Science in Liberal Arts & Sciences Major in Statistics
  - GPA: 3.99/4.00

Papers

SUB=Submitted, ACC=Accepted

2025 **[ACC] RSK linear operators and the Vershik-Kerov-Logan-Shepp curve** [arXiv]
Duy Phan, **David Xia**
to appear in Electronic Journal of Combinatorics

2024 **[ACC] EveGuard: Defeating Vibration-based Side-Channel Eavesdropping with Audio Adversarial Perturbations** [arXiv]
Jung-Woo Chang, Ke Sun, **David Xia**, Xinyu Zhang, Farinaz Koushanfar
IEEE Symposium on Security and Privacy, 2025

Workshop Papers & Manuscripts

PRE= In Preparation, SUB=Submitted, ACC=Accepted

2026 **[PRE] Frozen Policy Iteration for MDPs with Stochastic Transitions under Linear Q^π Realizability** [PDF]
David Xia, Ruizhong Qiu, Hanghang Tong
Manuscript in Preparation

2026 **[SUB] Predictability in Major League Sports: Betting Odds versus Mathematical Models** [PDF]
David Xia, AJ Hildebrand
Submitted to the Mathematics and Sports (MAS) Journal

2026 **[ACC] Revitalizing Local Democracy: A Human-Centered Audit of LLMs in City Council Journalism** [PDF]
David Xia, Chris Maury
Human-centered Evaluation and Auditing of Language Models (HEAL) Workshop at CHI 2026

Research Experience

- **Northwestern University**, Research Intern *May 2026 - present*
  - Advised by Professor Zhaoran Wang.
  - Developed a closed-loop LLM meta-orchestrator agent for aircraft wing inverse design, where the agent diagnoses its own surrogate failures across iterative versions and emits corrective source code for the next optimization layer.
  - Identified and systematically addressed seven structural failure modes of surrogate-based optimization—including surrogate exploitation, Gaussian trust collapse, family-specific exploitation, and single-objective specification gaming—building a unified taxonomy and corresponding defenses.
  - Applied reinforcement-learning-style reasoning to the outer loop: the LLM agent classifies failure modes and upgrades its decision layer, achieving relative robustness gains validated against public RANS datasets (AirfRANS, SuperWing, ONERA CRM).
- **iDEA-iSAIL Lab**, Research Intern *Feb 2026 - present*

- Advised by Professor Hanghang Tong.
- Conducted research in theoretical reinforcement learning, studying sample and computation efficient algorithms for online learning in finite-horizon Markov Decision Processes under function approximation and stochastic dynamics.
- Used concentrability to develop the first sample and computation efficient algorithm for online reinforcement learning in stochastic-transition MDPs under the linear Q^π realizability assumption, resolving an open problem left by prior work restricted to deterministic dynamics.
- Established a PAC sample complexity guarantee of $\tilde{O}(d^2 H^7 C^* / \varepsilon^3)$ by combining techniques from both offline and online reinforcement learning theory, dynamic programming, concentration analysis, and linear function approximation.
- Manuscript in preparation.

- **Illinois Combinatorics Lab for Undergraduate Experience (ICLUE)**, Research Intern *Aug 2024 - present*
  - Advised by Professor Alexander Yong.
  - Proved a new asymptotic result that the probability that the Schensted insertion algorithm for a uniformly random permutation in S_n exhibits a special "bumping" interaction converges to 1 as n approaches infinity.
  - Derived the result using rigorous probability bounds on the RSK linear operator by leveraging the Vershik-Kerov-Logan-Shepp limit shape curve as the central analytic tool.
  - Developed proofs requiring techniques spanning real analysis (uniform convergence, limit arguments), probability theory (concentration inequalities, convergence in probability), and asymptotic analysis (bounding combinatorial quantities in the large-n regime).
  - Paper in submission to the Electronic Journal of Combinatorics.

- **University of California San Diego**, Research Intern *Jun 2023 - Dec 2024*
  - Advised by Professor Xinyu Zhang.
  - Developed EveGuard, a software-based defense framework to protect voice privacy from vibrometry-based side channels using adversarial audio by using a perturbation generator model (PGM) to effectively suppress sensor-based eavesdropping while preserving high audio quality.
  - Implemented Eve-GAN, a novel domain translation task for inferring eavesdropped signals, enabling end-to-end training of PGM, utilizing few-shot learning techniques to reduce data collection overhead.
  - Achieved a protection rate of over 97% against audio classifiers, hindering eavesdropped reconstruction.
  - Paper published in the IEEE Symposium on Security and Privacy 2025.

- **Illinois Mathematics Lab (formerly Illinois Geometry Lab)**, Research Intern *Jan 2024 - present*
  - Advised by Professor AJ Hildebrand.
  - Researched the data science perspective of sports matches, analyzing the predictability of outcomes by comparing prediction accuracies across polls, the betting market, Elo ratings, and mathematical models like Bradley-Terry.
  - Presented findings at Rose-Hulman Undergraduate Math Conference 2024, UIUC Undergraduate Research Symposium 2024, and Joint Mathematics Meeting 2025.
  - Paper in submission to the Mathematics and Sports (MAS) Journal.

- **Carnegie Mellon University Human-Computer Interaction Institute**, Research Intern *May 2025 - Feb 2026*
  - Advised by Professor Jeff Bigham.
  - Designed and evaluated an automated journalism pipeline using SOTA LLMs to replicate end-to-end editorial workflows, from transcript segmentation to headline generation and topic prioritization.
  - Orchestrated large-scale crowd-sourced study and demonstrated that LLM-based headline quality and topic prioritization can meet and exceed professional standards.
  - Paper in CHI 2026 Human-centered Evaluation and Auditing of Language Models (HEAL) workshop.

Relevant Coursework

CS: Statistical Reinforcement Learning, Machine Learning, NLP, Deep Learning, Algorithms, Numerical Methods, Numerical Analysis
Math: Honors Real Analysis, Graph Theory, Combinatorics, Abstract Linear Algebra, Abstract Algebra I–II, Algebraic Combinatorics, Linear Programming
Stats: Stochastic Processes, Applied Bayesian Analysis, Survival Analysis

Grants and Awards

2025 CMU HCII REU: NSF 2349558

2025 ICLUE RTG: NSF 1937241

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